WHOLE NUMBERS

SUBTOPIC: FORMING NUMERALS USING GIVEN DIGITS

CONTENT: Using digits to form smallest and largest numbers.
Finding the sum of/product/difference/quotient between the smallest and largest numbers formed from the given digits.

Examples:

- Write down all 3-digit numerals that can be formed using the digits; 4, 6, 7
  476 467 647 674 746 764
- Find the difference between the smallest and highest numerals formed.
  Smallest = 467
  Biggest = 764
  Difference =

$$\begin{array}{r} 7\,6\,4 \\ -\,4\,6\,7 \\ \hline 2\,9\,7 \\ \hline \end{array}$$

Activity:

- ✓ Using digits **3, 0, 6 ,**form all 3 - digit numerals that can be formed.
- ✓ Find the product of the smallest and the biggest numerals formed.

- ❖ Write down all three digit even numerals that can be formed from ***8, 3, 4.***
- ❖ Find the sum of all 3-digit odd numbers that can be formed

SUBTOPIC: PLACE VALUES AND VALUES OF DIGITS UP TO MILLIONS

The place value chart

Million			Thousands			Units		
H	T	O	H	T	O	H	T	O
		7	6	5	4	3	2	9

The table below shows the place values and values of the numeral above

$Digit$	$Place\ value\ in\ words$	$Place\ value\ in\ figures$	$Value\ (digit \times P.V)$
9	Ones	1	$9 \times 1 = 9$
2	Tens	10	$2 \times 10 = 20$
3	Hundreds	100	$3 \times 100 = 300$
4	Thousands	1,000	$4 \times 1{,}000 = 4{,}000$
5	Ten thousands	10,000	$5 \times 10{,}000 = 50{,}000$
6	Hundred thousands	100,000	$6 \times 100{,}00 = 600{,}000$
7	Millions	1,000,000	$7 \times 1{,}000{,}000 = 7{,}000{,}000$

Example: Find the value of 6 in the number 2602

2 6 0 2

- 2 → Thousands (1000)
- 6 → Hundreds (100)
- 0 → Tens (10)
- 2 → One (1)

Value of 6 = 6 x 100 = 600

Activity:

1. Write the place value and value of 9 in
   (a). 345987
   (b).26490321
   (c).689458345
2. Find the sum of the value of seven and the place value of 9 in 23745893

SUB TOPIC: WRITING IN WORDS (UP TO MILLION)

Examples:

(i) Write 20,480 in words.

Thousand	Units
20	480

Twenty thousand, four hundred eighty.

(ii) 6,808,040

Million	Thousand	Units
6	808	040

Six million, eight hundred eight thousand forty.

Activity:

Write the following in words.

- 34567
- 9999999
- 3,230,203
- 67,045
- 2,999,087

SUB TOPIC: WRITING NUMERALS IN FIGURES

Examples:

Write in figures:

(i) seven million, four hundred twenty one thousand, nine hundred five.

7 million = 7,000,000

421 thousand + 421,000

905 905

7,421,905

(ii) A quarter of a million

A million = 1,000,0000

$$\frac{1}{4}\ of\ 1,000,000$$

$$\frac{1}{4} \times 1,000,000$$

$$\underline{\underline{=\ 250,000}}$$

(iii) Write " six hundred thirty nine thousand, seven" in figures

(iv) Write " three million, thirty nine thousand, eight" in figures

(v) Write " nine million, four hundred thirty nine thousand, six" in figures

(vi) Write "thirty nine thousand, fifty seven" in figures

SUB TOPIC: EXPANDED NOTATION

CONTENT: Expanding numerals using:

- **Place values**
- **Values**
- **Powers of ten/exponents**

Examples:

Expand: 5624 using:

<u>Place values:</u> **5624 = (5 x 1000) + (6 x 100) + (2 x 10) + (4 x 1)**

<u>Values:</u> **5624 = 5000 + 600 + 20 + 4**

<u>Powers:</u>

10^3	10^2	10^1	10^0
5	6	2	4

5624 = $(5 \times 10^3) + (6 \times 10^2) + (2 \times 10^1) + (4 \times 10^0)$

Activity:
Expand the following as instructed
a) 2,354 (place values)
b) 40,369 (place values)
c) 45,689 (values)
d) 29,542(values)
e) 890765 (exponents)
f) 2354 (powers)

SUB TOPIC: FINDING THE EXPANDED NUMBERS (SHORT FORM)

Examples:
Write as a single number.

(i) (6 x 10,000 + (4 x 10) + (5 x 1)
(6 x 10000) + (4 x 100) + (5 x 1)
= 60,000 + 400 + 5
= 60000
400
+ 5
60405

(ii) 9000000 + 700 00 + 50000 + 1000 + 30 + 8
= 9 000 000
700 000
50 000
1 000
30
+ 8
9 750 038

(iii) $(2 \times 10^5) + (4 \times 10^3) + (6 \times 10^0) + (7 \times 10^2)$
(2 x 10 x 10 x 10 x 10 x 10) + (4 x 10 x 10 x 10) + (6 x 1) + (7 x 10 x 10)
200,000 + 4000 + 6 + 700
200 000
4000
700
+ 6
204706

<u>Activity</u>

What number has been expanded to give?

a) $3000 + 200 + 3$

b) $(5 \times 1000) + (9 \times 100) + (2 \times 10) + (8 \times 1)$

c) $(9 \times 10^4) + (9 \times 10^2) + (2 \times 10^1) + (8 \times 10^0)$

SUB TOPIC: ROUNDING OFF WHOLE NUMBERS

Examples:

Round off the following as instructed.

(i) 3864 to the nearest hundreds.

TH	*H*	*T*	*0*
3	**8**	**~~6~~**	**~~4~~**
+	**1**		
3	**9**	**0**	**0**

(ii) 214 (nearest tens

H	*T*	*O*
2	**1**	**~~4~~**
+	**0**	
2	**1**	**0**

(iii) 4.78516 to the nearest thousandths.

0	**T^{th}**	**H^{th}**	**TH^{th}**	**T/TH^{th}**
4 .	**7**	**8**	**5**	**~~1~~**
+ 0 .	**0**	**0**	**0**	
4.	**7**	**8**	**5**	

(i) 75.634 to the nearest whole number nearest whole number

T	**0**	**Tth**	**Hth**	**THth**
7	**5 .**	**~~6~~**	**~~3~~**	**~~4~~**
+	**1**			
7	**6 .**			

Activity:

1. **Round off the following as instructed in brackets**
   a) **45637 (nearest hundreds)**
   b) **99999 (nearest thousands)**
   c) **780937887 (nearest millions)**
   d) **89.58(nearest tenths)**
   e) **23.786(nearest hundredths)**
   f) **7.239(nearest two d.p)**

SUB TOPIC: ROMAN NUMERALS (converting Hindu Arabic numerals to roman numerals)

BASIC ROMAN NUMERALS

1 =I
5 =V
10 = X
50 =L
100 =C
500 =D

NOTE:
All other numerals are got from basic roman numerals by adding or subtracting .

Example

1. Write the following in roman numerals

1 2 4

100 + 20 + 4
100 = C
20 = XX
4 = IV
∴ 124 = ***CXXIV***

ii) 1962
1000 + 900 + 60 +2
1000 = M
900 = CM
60 = LX
2 = II
∴ 1962= ***MCMLXII***

Activity

Write the following in roman numerals

a) *49*
b) *235*
c) *333*
d) *78*
e) *140*
f) *999*
g) *1449*

SUBTOPIC:ROMAN NUMERALS (conversion of Roman numerals to Hindu Arabic)

Write the following numbers in Hindu Arabic numerals

(i) MXLV

M + XL + V
M = 1000
XL = 40
V = 5
MXLV = 1045

(ii) CD XCIV

CD + XC + IV

CD = 400

XC =   90

IV =   4

CDXCIV = 494

(iii)   Write the following in Hindu Arabic numerals

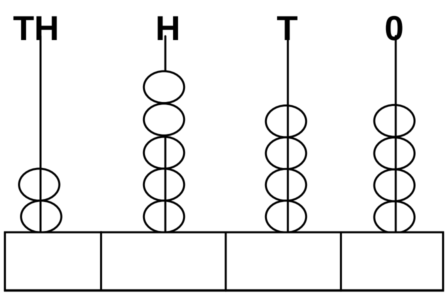
- *XLIV*
- *CVIII*
- *XCIX*
- *CDXCIV*
- *MXCIV*

(iv)   A church was built in MDCCLXIV. Which year is this in Hindu Arabic?

END OF TOPIC EXERCISE

1. Show 2843 on the abacus.

2. Write the number shown on the abacus in roman numerals.



3. Write the place value of each digit in 486349334.

4. Workout the sum of the value of 6, 8 and 7 in 4638047.

5. Workout the difference between that value of 7 and the place value of 9 in 49834734.

6. Write the product of the value of 6 and 3 in 8469 in roman numerals.

7.   Write 4009009 in words.

8.   Write “Two hundred thirty six thousand forty nine” in expanded form using exponents.

9.   Expand 963.07 using values.

10. Which number was expanded to give $(6x10^4) + (8 \times 10^2) +(3 \times 10^{-1}) + (7x10^{-2})$

11. Round off 9999 to the nearest thousands.

12. Round off 67.987 to the nearest hundredths.

13. Round off 99.999 to the nearest hundredth.

14. Round off 16.873 to the nearest whole number.

15.   Expand 635.057 using exponents.

16.   St. Mary’s college was constructed in MCMLXXXVII. In which year was it constructed?

17.   Write 18.487 in words.

18.   Find the sum of the value of 7 and the value of 3 in 3.467.

19. Write forty six and eighty seven thousandths in figures.